

FPGA-based Acceleration for Convolutional Neural Networks: A Comprehensive Review

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Abstract—Convolutional Neural Networks (CNNs) are fundamental to deep learning, driving applications across various domains. However, their growing complexity has significantly increased computational demands, necessitating efficient hardware accelerators. Field-Programmable Gate Arrays (FPGAs) have emerged as a leading solution, offering reconfigurability, parallelism, and energy efficiency. This paper provides a comprehensive review of FPGA-based hardware accelerators specifically designed for CNNs. It presents and summarizes the performance evaluation framework grounded in existing studies and explores key optimization strategies, such as parallel computing, dataflow optimization, and hardware-software co-design. It also compares various FPGA architectures in terms of latency, throughput, compute efficiency, power consumption, and resource utilization. Finally, the paper highlights future challenges and opportunities, emphasizing the potential for continued innovation in this field.

Index Terms—Convolutional Neural Networks (CNNs), Field Programmable Gate Arrays (FPGAs), Hardware Accelerator, Compute Efficiency, Parallel Computing, Design Space Exploration (DSE)

I. INTRODUCTION

In recent years, Deep Learning (DL) has gained significant attention in computer science [1], with Convolutional Neural Networks (CNNs) serving as a cornerstone of this progress. CNNs have achieved remarkable success across various applications, including image classification [2], object detection [3], and semantic segmentation [4]. Their transformative impact has led to widespread adoption by industry leaders, recognizing CNNs as a vital tool for innovation and competitive advantage.

The strength of CNNs lies in their ability to process high-dimensional data through convolutional operations, pooling, and non-linear activations, enabling automated multi-level feature extraction. However, this capability comes at the cost of increased computational complexity and resource demands. With their adoption in mobile and edge devices like drones and smartphones [5], CNNs face critical challenges in achieving high performance under resource constraints.

To address these challenges, extensive research has been conducted on hardware acceleration for CNNs [6]–[9]. While CPUs, GPUs and ASICs are commonly employed, their limitations in handling the complexity of modern CNN architectures

and real-time processing needs have prompted growing interest in alternative platforms. Field-Programmable Gate Arrays (FPGAs) have emerged as a promising solution, offering reconfigurability, parallel processing, and energy efficiency. Their ability to customize hardware for specific tasks enables the design of high-performance, resource-efficient accelerators tailored for deep learning applications.

This review article explores recent advances in FPGA-based CNN accelerators, focusing on acceleration methods, architectural innovations, hardware optimization techniques, and software-hardware co-design frameworks. It highlights key trends, addresses current challenges, and outlines future directions in this rapidly evolving domain. Additionally, the article examines performance evaluation metrics and provides insights into diverse architectures, offering practical guidance for researchers and practitioners. The main contributions of this work are summarized as follows:

- **Evaluation Metrics:** We review FPGA-based CNN acceleration research, consolidating key performance metrics, particularly compute and overall efficiency. It explores the relationships between these metrics, aiding researchers in identifying design limitations and guiding future advancements (Section III);
- **Acceleration Methods:** We present a quantitative comparison of existing CNN acceleration methods, providing practitioners and researchers with valuable insights to select approaches that best align with their specific system requirements (Section IV);
- **Parallel Computing Analysis:** This study reviews parallel computing methods and their hardware architectures in existing designs, analyzing their strengths and weaknesses using the introduced evaluation metrics. It highlights the importance of dynamic parallelism in enhancing hardware performance, including compute efficiency and resource utilization (Section V);
- **Future Directions:** Building on the comprehensive review in this work, we propose and discuss future research directions to help researchers and designers identify key design considerations and potential optimizations in this field (Section VII).

The rest part of this paper is organized as follows: Section II compares the acceleration platforms for CNNs, emphasizing the advantages of FPGAs. In Section III, we analyze and outline design performance metrics for FPGA-based CNN accelerators, guiding the comparison of existing designs in subsequent sections. This section also highlights commonly overlooked metrics in the literature and suggests directions

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